Fiscal Year:	FY 2004	**Task Last Updated:**	FY 05/18/2004
PI Name:	Goodstein, David L. Ph.D.		
Project Title:	The CQ Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	91125	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Original end date was 4/30/2007		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Duncan, Robert (University of New Mexico) Chui, Talso (JPL) Day, Peter (JPL)		
Grant/Contract No.:	NAG 3 2900		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

	The heat capacity of superfluid helium in a constant heat flux has been predicted to become infinite at a critical value of the heat flux where superflow becomes unstable. Measurements done in our laboratory at Caltech have succeeded in detecting an excess heat capacity due to the presence of a heat flux but, for a variety of gravity-related reasons, it is impossible in the laboratory to make measurements at values of the heat flux that approach the critical value. The CQ Experiment will measure the heat capacity at constant heat flux in the absence of gravity. The CQ Experiment will be a guest experiment, using the hardware of Critical Dynamics in Microgravity (DYNAMX) without engineering modification. The DYNAMX Instrument Sensor Package (ISP) was not designed for precision calorimetry, but is nearly ideal for that purpose. Scheduled for the M1 mission aboard the Low Temperature Microgravity Physics Facility (LTMPF), DYNAMX should require about one month to complete its measurements. The liquid helium on the M1 mission is expected to last 4-5 months. The CQ experiment will be done in the DYNAMX ISP during the remaining time. The CQ Experiment will measure the heat capacity of superfluid helium at constant heat flux, over a range of temperatures within one millionth of a Kelvin of the superfluid transition (the lambda point, where helium becomes superfluid in the absence of a heat flux). The range of heat fluxes will be from zero up to a few millionths of a Watt per square centimeter. Only in these conditions can helium conduct heat at constant temperature (within the 50-picokelvin resolution of our thermometry). At higher heat fluxes, temperature gradients would render heat capacity measurements meaningless. With these small heat fluxes the experimental signal, excess heat capacity due to the heat flux, becomes resolvable only within one microkelvin of the lambda point.
Task Description:	The goal of the experiment is to measure the heat capacity as close as possible to the critical value of the heat flux, where superflow becomes unstable, and where the heat capacity is predicted to approach infinity. On Earth, the hydrostatic pressure of a column of liquid helium causes the lambda point temperature itself to vary by more than one microkelvin per centimeter of column height. To combat this problem, our laboratory measurements were made in a cell only 0.06 centimeters high. Even so, all our data are severely compromised by the hydrostatic gravity effect. In addition, the very small cell height made it necessary to rely on temperatures measured in the end caps of the cell, greatly complicating data analysis and limiting the range of measurements to values of the heat flux far below the critical value. All or most of these problems will be overcome by making the measurements in the DYNAMX ISP. The DYNAMX cell is 11 mm high and measures the temperature with high resolution thermometry at three positions along the side wall. There are two outstanding discrepancies in this field at the moment. The first is that experiment and theory disagree on the value of the critical heat current as a function of temperature, and the other is that the excess heat capacity measured in our own experiments is much larger than that predicted theoretically. A successful CQ experiment would shed considerable light on the causes of both of these discrepancies. If in addition it turns out to be possible to measure the heat capacity close enough to the critical heat current to observe the functional form of the predicted divergence, that result might very well open a whole new field of physics. The form of the divergence should be dominated by fluctuations of a qualitatively different kind from those that govern the behavior of conventional critical points.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Scientific progress: During FY04 we have continued to explore the Self Organized Critical (SOC) state, an exotic state of matter that can also exist in liquid helium. As before this work has been done in conjunction with our collaborators at the University of New Mexico, under the DYNAMX project. SOC states exist throughout nature: In this state the system, though subject to external forces that would normally drive it away from equilibrium, manages to exist in a steady-state close to its phase transition, or “critical line”. In our case the system is liquid helium, the critical line is the superfluid transition (T_lam) and the external force is the heat flux (Q) applied to the helium sample. As the helium enters this state each point in the sample ‘self-organizes’ to have the same temperature difference from T_lam. Expanding on our previous measurements of the heat capacity of liquid helium as it transitions into the SOC state, the CQ project is conducting new experiments at Caltech to measure the heat capacity of the full SOC state. At the same time, in collaboration with DYNAMX, we have designed an improved experimental cell that will be used to make new measurements of CQ – the heat capacity under a constant heat flux. This new cell will take advantage of improvements in low temperature thermometry and construction techniques (some of which were a direct consequence of the DYNAMX/CQ project) to re-confirm our previous ground-based measurements, and may help to clarify some of the discrepancies between those previous measurements and theory. Once these new measurements have been made they may very well represent just about the best that can be achieved on Earth under the deleterious effect of gravity. So far in FY04 this work has lead to one invited talk and two published papers. Project progress: During FY04 CQ’s main emphasis has been the development of a flight simulator to aid our software development effort. A simple version of this simulator has already been created at Caltech and, while being primitive, has proven to be a valuable test bed for developing some of the CQ algorithms in “Python” - the chosen flight software language. Due to the termination of the CQ flight project (in September 04) no further development of the full flight simulator will take place. Instead, future work (in FY04) will concentrate on further development of the CQ algorithms using the Caltech basic simulator and on additional ground experimentation at UNM/Caltech, using NASA/JPL hardware and, were applicable, expert advice from the JPL staff in the Low Temperature Engineering Division.
Bibliography Type:	Description: (Last Updated: 02/25/2014)
Articles in Peer-reviewed Journals	Sergatskov DA, Babkin AV, Boyd STP, Lee RAM, Duncan RV. "Experiments with He-4 heated from above, very near the lambda point." Journal of Low Temperature Physics, 2004 Jan;134(1-2):517-26. , Jan-2004

Articles in Peer-reviewed Journals	Lee RAM, Chatto AR, Sergatskov DA, Babkin AV, Boyd STP, Churilov AM, McCarson TD, Chui TCP, Day PK, Duncan RV, Goodstein DL. "“Heat from above“ heat capacity measurements in liquid He-4." Journal of Low Temperature Physics, 2004 Jan;134(1-2):495-505. , Jan-2004
Presentation	Goodstein, D. L.; "A Dynamic New Look at the Lambda Transition" Condensed matter physics seminar, Physics Department, University of Alberta, Edmonton. Jan-2004